# Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

## 1 Tomato

### 1.1 Harvesting

- **Harvest maturity indices:** Harvest readiness in tomato (*Solanum lycopersicum*) is assessed primarily through fruit maturity stages and external colour progression, specifically identifying mature green and breaker stages. A standard documented indicator for fresh market tomatoes recommends harvesting mature green fruits when approximately 10% of the fruits on the first hand reach the breaker stage. In processing hybrids such as Arka Apeksha and Arka Vishesh, vine storability evaluations demonstrated that fruits remain intact on plants up to 110 days after transplanting, allowing once-over harvest readiness. Documented physical characteristics associated with harvested fruits include pericarp thickness ranging from 0.30–0.97 cm and fruit firmness values between 6.00–10.33 ($kg/cm^2$ ).$Specificphysiologicalmaturitythresholds, exactfruitdimensions, anddaysafterfloweringindi$

- **Harvesting time and schedule:** Picking schedules are strongly dictated by ambient weather conditions, requiring tomato fruits to be harvested at 2–3 day intervals during warm weather and at weekly intervals during cool weather. Vine-ripened tomatoes must be hand harvested every other day. Total crop duration varies by cultivar, documented at 165 days for Dhanshree and Phule Jayshree, and 180 days for Phule Kesari. Sowing and planting schedules span multiple seasonal windows, including May–June sowing with June–July planting, September–October sowing with October–November planting, and December–January sowing with January–February planting. First harvest timing, exact number of pickings, specific time of day for harvesting, and daily picking schedules are Not specified in the provided sources.

- **Harvesting methods:** Harvesting is conducted manually by twisting the fruits rather than pulling them from the plants to prevent mechanical damage to the vine and fruit. Fruits are generally removed from the upper portion of the plant. Processing hybrids Arka Apeksha and Arka Vishesh are bred with jointless pedicels (*j2*), which facilitate rapid hand picking or mechanical harvesting and allow a once-over harvest even 110 days after transplanting. Staking plants improves fruit quality and ease of picking by keeping vines off the ground. Specific harvesting tools, collection machinery, and whole-plant harvesting procedures are Not specified in the provided sources.

- **Fruit maturity stages:** Documented maturity classifications explicitly include immature green, mature green, breaker, and vine-ripe or red-ripe stages. Immature green fruits possess lower sugar and acid contents, which severely reduces

eating quality even if artificially gassed to redness. Mature green fruits develop full flavour comparable to vine-attached fruit within 24 hours of colour appearance. The breaker stage is characterized by the initial appearance of colour on the fruit. Specific descriptions for turning, pink, and light red stages are Not specified in the provided sources.

- **Harvesting by intended use:** Maturity stages are strictly aligned with market destinations and end uses. Commercial fresh market crops intended for distant transportation are harvested at the mature green stage, specifically when about 10% of fruits on the first hand reach the breaker stage, allowing long-distance transport prior to artificial ripening. Fresh market tomatoes intended for direct or local sales are harvested vine-ripe every other day. Cultivars intended for industrial processing, such as Arka Apeksha, Arka Vishesh, and Dhanshree, require high yield, jointless pedicels for once-over or rapid harvesting, high total soluble solids of 5–6
(°Brix
)($or 4.5 - -5.4$
(°Brix
)), $acidity not less than 0.4\%, pH below 4.5, and uniform dark red colour with ana/b colour value of at le$

- **Harvesting precautions:** To prevent mechanical injury, rot, and quality loss, tomatoes must be twisted off plants instead of pulled, avoiding stem and vine damage. Vines and fruits must be completely dry when harvesting mature green tomatoes; harvesting wet fruits causes sunken, blackened areas to develop during subsequent ripening. Chemical pesticide application waiting periods must be strictly observed prior to picking. In post-harvest water cleaning tanks, fruit must not be submerged deeply or float more than one layer deep to prevent water pressure from forcing pathogen-containing water into the stem scar.

- **Labour and harvesting equipment:** Harvesting efficiency is enhanced in processing cultivars containing jointless pedicels ($j2$), which enable mechanical harvesting equipment or rapid hand picking by field labour. Field collection systems utilize water tanks for cleaning and disease management during post-harvest handling. Specific numerical labour requirements, picking containers, field crates, harvesting tool specifications, and mechanical harvester equipment details are Not specified in the provided sources.

- **Yield and productivity:** Average tomato productivity in India is reported at 24 t/ha across an estimated 0.76 million hectares. Processing hybrids Arka Apeksha and Arka Vishesh exhibit high yield potentials of 80–90 t/ha (with estimated experimental yields ranging from 24.67–43.67 t/ha and 28.07–52.71 t/ha depending on season and trial). Documented yields for other university-developed varieties include 40–45 t/ha for Dhanshree, 55–60 t/ha for Phule Kesari, and 50 t/ha for cherry tomato Phule Jayshree. Average fruit weight varies by cultivar, documented at 66.67–125.00 g for Arka Apeksha and 68.70–105.30 g for Arka Vishesh, with 8.00–15.67 fruits/kg. Factors influencing harvested yield include staking, balanced fertigation, plant growth regulators such as PCPA foliar sprays at 50–100 mg/L during extreme flowering temperatures, and Gibberellic acid sprays at 10–20 mg/L during low temperatures.

- **Fruit sorting and grading:** Sorting and grading criteria focus on fruit firmness, pericarp thickness, shape, size, color value, and freedom from physiological defects or disease symptoms. Marketable fruits must be free from defects such as blossom-end rot, cat-facing, radial and concentric cracking, gray wall (blotchy ripening), zebra stripe, and zippering with open holes. For industrial processing quality, sorting standards mandate total soluble solids of 5–6
(°Brix
), $acidity not less than 0.4\%, pH less than 4.5, and uniform red color index. Standardized commercials$

- **Post-harvest handling:** Post-harvest handling procedures comprise field collection, deleafing, water-tank washing, precooling, sorting, grading, bunching, packaging, temporary storage, and preparation for transport. Fresh market tomatoes are dropped into cleaning water tanks where disease spread is controlled by restricting submersion depth, keeping fruit no deeper than one layer, limiting washing duration to 2 minutes, slightly chlorinating the water, and warming the tank water a few degrees above the fruit temperature to prevent cool water from constricting fruit and drawing pathogens into the stem scar.

- **Packaging and transportation:** Post-harvest packaging standards and infrastructure are required to enhance holding life and reduce losses across export, distant, and local market supply chains. Packing tomatoes immediately following harvest and cleaning prepares them for fresh market transit. Specific packaging materials, crate dimensions, container types, loading techniques, stacking limits, and transport temperature or ventilation specifications are Not specified in the provided sources.

- **Storage conditions and shelf life:** Storage duration depends directly on the fruit maturity stage at harvest. Mature green fruits can be stored for up to 30 days under cool storage conditions at
(10°C
). $Fully ripe fruits can be stored for approximately 1 week. Vine storability on intact plants in the fielde$

- **Ripening and ethylene management:** Artificial uniform ripening of commercial fresh market tomatoes harvested at the mature green stage is achieved by dipping fruits in ethephon 39% SL at 2500 ppm. Controlled pre-harvest fruit ripening is induced by foliar spraying ethephon at 1000 mg/L (1000 ppm). Immature green tomatoes can be artificially gassed to redness using ethylene, but this fails to increase sugar and acid levels, yielding flavourless fruit. Naturally ripened mature green tomatoes achieve full flavour within 24 hours of colour appearance.

- **Post-harvest losses and quality deterioration:** Causes of post-harvest losses and fruit rejection include mechanical injury from pulling or harvesting wet vines, stem scar pathogen ingress during improper water tank washing, and physiological disorders. Blossom-end rot causes localized calcium deficiency leading to light tan water-soaked lesions that turn black and leathery. Cat-face causes severe blossom-end deformities due to low temperatures during flowering, herbicide drift, or thrips feeding. Fruit cracking occurs when internal expansion exceeds epidermal expansion, triggered by sudden heavy watering following dry periods. Gray wall (blotchy

ripening) causes internal dark necrotic vascular tissue and grayish outer wall collapse, exacerbated by cool short days and excess nitrogen. Zippering causes longitudinal scars with open locular holes.

- **Food safety and sanitation:** Sanitation during post-harvest handling requires strict water tank hygiene to prevent disease transmission. Water tanks must be slightly chlorinated, maintained slightly warmer than fruit temperatures, and kept shallow so that pressure does not force contaminated water through the stem scar. Pesticide application safety waiting periods must be strictly observed prior to harvest. Sanitizer concentrations, equipment disinfection protocols, and formal food safety certifications are Not specified in the provided sources.

- **Effects on fruit quality and marketability:** Harvesting mature green fruits ensures extended storage shelf life up to 30 days and permits long-distance transport, but harvesting at immature green stages irreversibly lowers sugar and acid content, impairing taste and market value. Staking vines improves fruit quality by preventing ground contact and reducing wetness. Proper water tank temperature and chlorination preserve fruit soundness by preventing decay losses. Industrial processing suitability depends on high fruit firmness (6.00–10.33 ($kg/cm^2$ )), $pericarp thickness (0.30 - -0.97 cm)$, $high total soluble solids (4.5 - -6$ (°Brix )), $and short processing pulp time (2 hours 12 minutes for Arka Apeksha compared to 2 hours 27 minute$

- **Scheduling and monitoring:** Field monitoring and scheduling involve tracking crop duration (100–180 days across varieties), scheduling harvest intervals (2–3 days in warm weather, 7 days in cool weather), and monitoring pest-to-defender ratios or AESA observations. Harvest record formats, formal yield logs, grading records, storage monitoring protocols, and loss assessment tracking are Not specified in the provided sources.

- **Integrated harvesting workflow:** The documented integrated workflow begins with pre-harvest care, pesticide waiting period verification, and maturity assessment (identifying 10% breaker stage on the first hand for mature green harvest). Harvesting is performed manually by twisting dry fruits or mechanically using jointless pedicel cultivars. Harvested fruits undergo field collection and water-tank cleaning using chlorinated, warmed water for 2 minutes to manage pathogens. Fruits are deleafed, sorted, graded against physiological defects, precooled, packed, and assigned to local mandis, processing facilities, or distant markets under cool storage ( (10°C )$for up to 30 days for mature green, or 1 week for ripe fruit) or artificial ripening using ethephon prior$